Then, $(\vec{a}\times\vec{b})\times(\vec{c}\times\vec{d}) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 4 & 4 & 0 \\ 8 & -2 & -6 \end{vmatrix} = -24\hat{i}+24\hat{j}-40\hat{k}$ ... (1)

On the other hand, we have

$$[\vec{a},\vec{b},\vec{d}]\vec{c}-[\vec{a},\vec{b},\vec{c}]\vec{d} = 28(3\hat{j}-\hat{k})-12(2\hat{i}+5\hat{j}+\hat{k}) = -24\hat{i}+24\hat{j}-40\hat{k} \quad \text{... (2)}$$

Therefore, from equations (1) and (2), identity (i) is verified.
The verification of identity (ii) is left as an exercise to the reader. ■

EXERCISE 6.3

1. If $\vec{a}=\hat{i}-2\hat{j}+3\hat{k}, \vec{b}=2\hat{i}+\hat{j}-2\hat{k}, \vec{c}=3\hat{i}+2\hat{j}+\hat{k}$, find (i) $(\vec{a}\times\vec{b})\times\vec{c}$ (ii) $\vec{a}\times(\vec{b}\times\vec{c})$.
2. For any vector $\vec{a}$, prove that $\hat{i}\times(\vec{a}\times\hat{i})+\hat{j}\times(\vec{a}\times\hat{j})+\hat{k}\times(\vec{a}\times\hat{k})=2\vec{a}$.
3. Prove that $[\vec{a}-\vec{b}, \vec{b}-\vec{c}, \vec{c}-\vec{a}]=0$.
4. If $\vec{a}=2\hat{i}+3\hat{j}-\hat{k}, \vec{b}=3\hat{i}+5\hat{j}+2\hat{k}, \vec{c}=-\hat{i}-2\hat{j}+3\hat{k}$, verify that
   (i) $(\vec{a}\times\vec{b})\times\vec{c}=(\vec{a}\cdot\vec{c})\vec{b}-(\vec{b}\cdot\vec{c})\vec{a}$ (ii) $\vec{a}\times(\vec{b}\times\vec{c})=(\vec{a}\cdot\vec{c})\vec{b}-(\vec{a}\cdot\vec{b})\vec{c}$
5. $\vec{a}=2\hat{i}+3\hat{j}-\hat{k}$, $\vec{b}=-\hat{i}+2\hat{j}-4\hat{k}$, $\vec{c}=\hat{i}+\hat{j}+\hat{k}$ then find the value of $(\vec{a}\times\vec{b})\cdot(\vec{a}\times\vec{c})$.
6. If $\vec{a},\vec{b},\vec{c},\vec{d}$ are coplanar vectors, show that $(\vec{a}\times\vec{b})\times(\vec{c}\times\vec{d})=\vec{0}$.
7. If $\vec{a}=\hat{i}+2\hat{j}+3\hat{k}, \vec{b}=2\hat{i}-\hat{j}+\hat{k}, \vec{c}=3\hat{i}+2\hat{j}+\hat{k}$ and $\vec{a}\times(\vec{b}\times\vec{c})=l\vec{a}+m\vec{b}+n\vec{c}$, find the values of l,m,n.
8. If $\hat{a},\hat{b},\hat{c}$ are three unit vectors such that $\hat{b}$ and $\hat{c}$ are non-parallel and $\hat{a}\times(\hat{b}\times\hat{c})=\frac{1}{2}\hat{b}$, find the angle between $\hat{a}$ and $\hat{c}$.

6.7 Application of Vectors to 3-Dimensional Geometry

Vectors provide an elegant approach to study straight lines and planes in three dimension. All straight lines and planes are subsets of $\mathbb{R}^3$. For brevity, we shall call a straight line simply as line. A plane is a surface which is understood as a set P of points in $\mathbb{R}^3$ such that , if $A, B,$ and C are any three non-collinear points of P, then the line passing through any two of them is a subset of P. Two planes are said to be intersecting if they have at least one point in common and at least one point which lies on one plane but not on the other. Two planes are said to be coincident if they have exactly the same points. Two planes are said to be parallel but not coincident if they have no point in common. Similarly, a straight line can be understood as the set of points common to two intersecting planes. In this section, we obtain vector and Cartesian equations of straight line and plane by applying vector methods. By a vector form of equation of a geometrical object, we mean an equation which is satisfied by the position vector of every point of the object. The equation may be a vector equation or a scalar equation.

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